ETHAN UPHOFF

MECHANICAL ENGINEERING PORTFOLIO

Mechanical Design | Hardware Development
Test & Analysis | Aerospace Systems

University of Nebraska-Lincoln

B.S. Mechanical Engineering | Robotics Minor

May 2027 | GPA 3.90 / 4.00

ENGINEERING EXPERIENCE

Cislune Inc. - Mechanical Systems & Testing Intern, 2026
Cislune Inc. - Mechanical Assistant, 2025
NASA Nebraska Space Grant - Undergraduate Researcher, 2025-2026

TOOLS

SolidWorks | Onshape | ANSYS
MATLAB | 3D Printing

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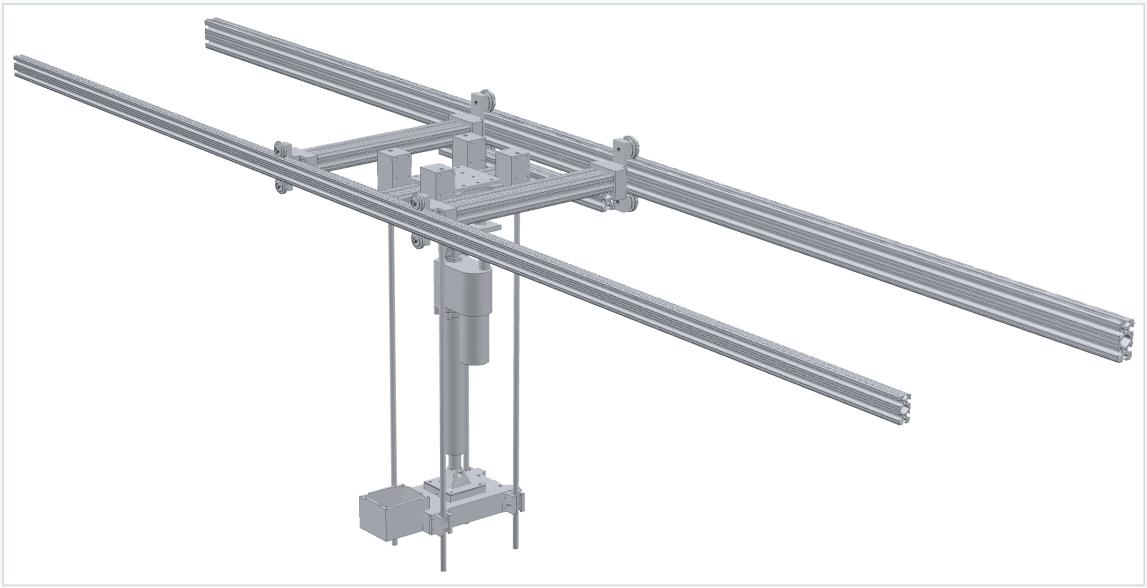
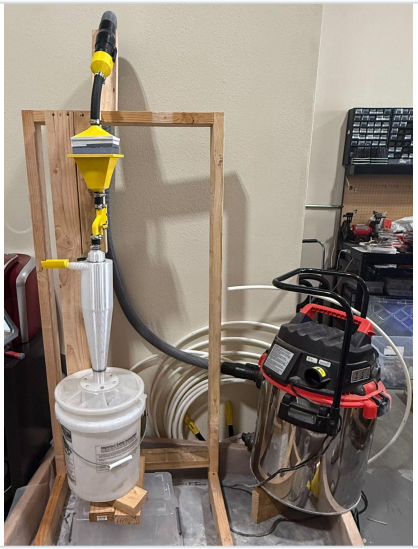
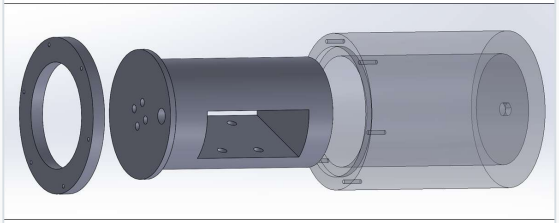


CRATR Mechanical test-system design & hardware development



SHEARPREP
Cyclone separation / CFD



VIBRATORY COMPACTION
Design / experimental research

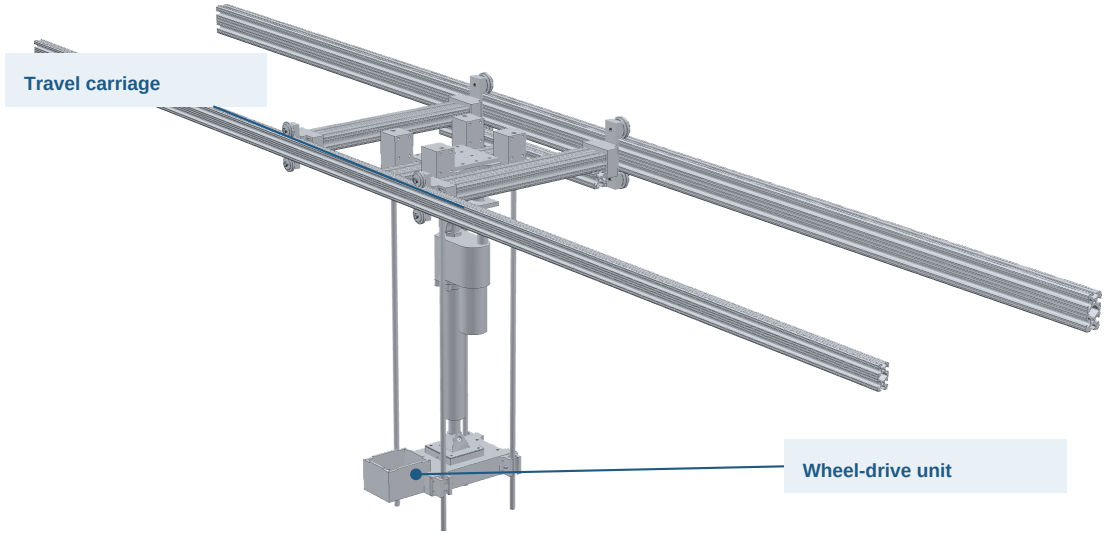
02

CRATR - Single-Wheel Regolith Test System

Cislune Inc. | Mechanical Systems & Testing Intern | Summer 2026

OUTCOME

Built a modular terramechanics rig that independently controls wheel motion and normal loading for repeatable rover wheel-regolith testing.



System-level CAD - 80/20 rail carriage, vertical load path and wheel-drive assembly.

Physical test hardware installed in the regolith bed.

OBJECTIVE

Create a repeatable single-wheel platform for comparing rover wheel designs in planetary regolith without requiring a full rover. The system needed controlled wheel speed, controlled vertical load and a modular test architecture.

MY CONTRIBUTION

Led the mechanical design and build. Designed the rail-mounted carriage, wheel-drive assembly and linear-actuator loading system in Onshape; rapidly prototyped custom components; sourced hardware within budget; assembled the system; and supported controls/electronics integration.

SYSTEM ARCHITECTURE

Horizontal translation and vertical loading were mechanically separated so wheel geometry, speed and applied load could be varied independently. A load cell measured the applied vertical force during testing.

8-16 in WHEEL RANGE	**5-15 rpm** TEST SPEED
20 N-m TORQUE TARGET	**8 in** REGOLITH DEPTH

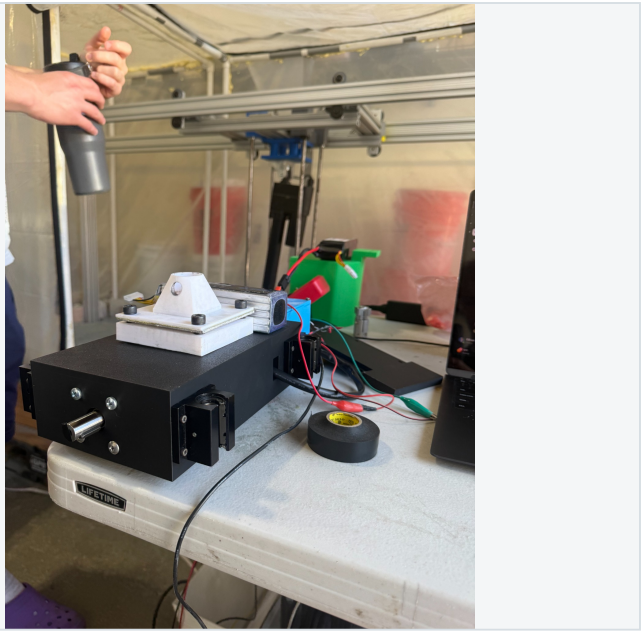
03

CRATR - Iteration Driven by Hardware Testing

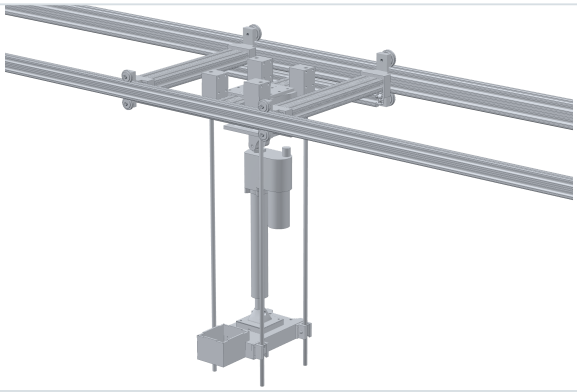
Failures in guidance, load alignment and the wheel mount directly informed redesigns.

OUTCOME

Testing exposed binding, friction, torque limits and repeated wheel-mount failures; each failure was converted into a specific mechanical design change.



Wheel-drive / motor box during integration.



Loading mechanism detail used to evaluate alignment and guidance.

ITERATION LOG

01

Carrier guidance

Lower carrier tilted on loose guide interfaces. Reworked guidance around higher-tolerance linear bearings.

02

Load-path alignment

Off-center actuator loading induced a moment and caused binding. Redesigned the lower carrier to center the applied load.

03

Friction & support

Guide friction limited low-load control. Cleaned/lubricated rods and improved upper guide support and clamping.

04

Powertrain & mount

Testing exposed torque limits and repeated printed mount failures. Increased robustness and replaced the critical shaft with metal hardware.

PHYSICAL TEST OUTCOME

20 lb @ 2 rpm -> 10 ft

After repeated mount failures, the metal-shaft revision completed the full 10 ft testbed travel without the prior shear failure.

04

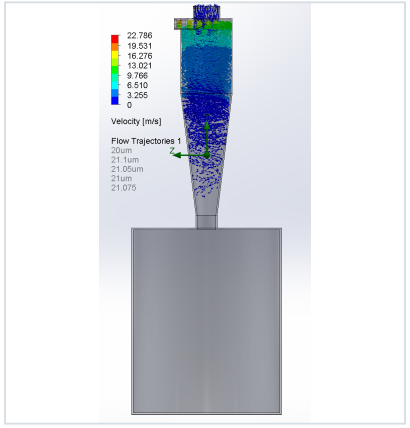
SHEARPREP - Cyclone Separation Development

Cislune Inc. | Mechanical Assistant | Summer 2025 | NASA STTR

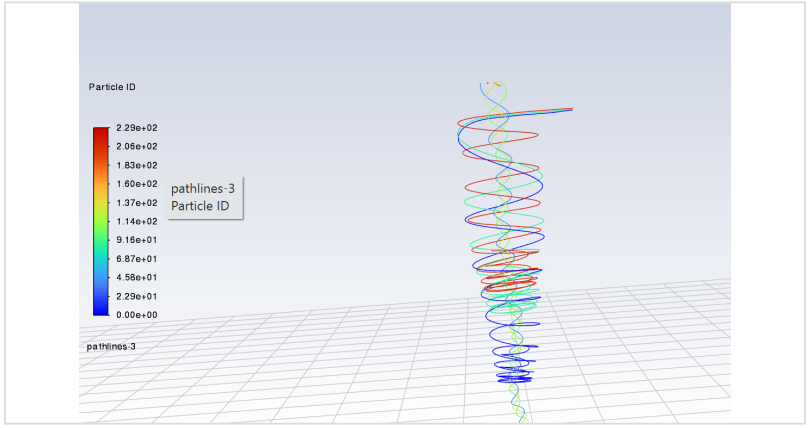
OUTCOME

Designed and prototyped cyclone-separation hardware, then paired bench testing with CFD to evaluate flow behavior and guide geometry iteration.

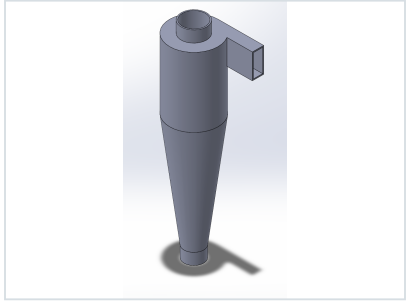
Bench-scale cyclone test stand.



Velocity field - SolidWorks Flow Simulation.



Particle-path analysis used to study separation behavior.



Cyclone CAD geometry.

ENGINEERING WORKFLOW

CAD > Prototype > Bench test > CFD > Iterate

Used physical observations and simulation together rather than treating CFD as a stand-alone result.

OBJECTIVE

Develop a pneumatic approach for classifying lunar regolith by particle size as part of the SHEARPREP surface-preparation system.

MY CONTRIBUTION

Designed and rapidly prototyped cyclone components in SolidWorks; supported test-rig construction, sealing and hardware iteration; and used SolidWorks Flow Simulation and ANSYS to investigate flow and particle behavior.

TECHNICAL FOCUS

Cyclone geometry, pneumatic transport, particle trajectories, repeatable test hardware and correlation between simulated and physical behavior.

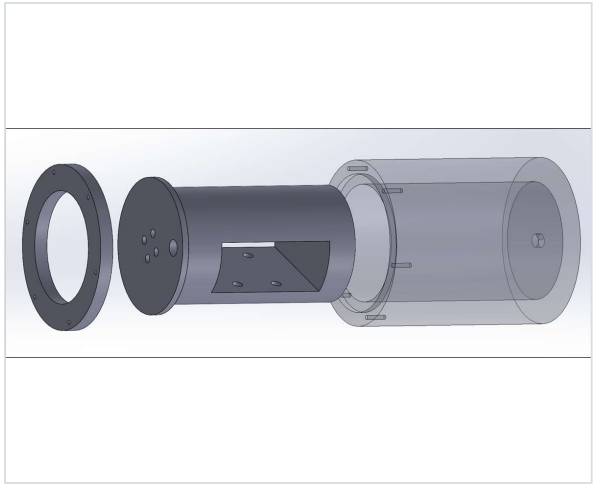
SolidWorks | ANSYS | CFD | 3D Printing

05

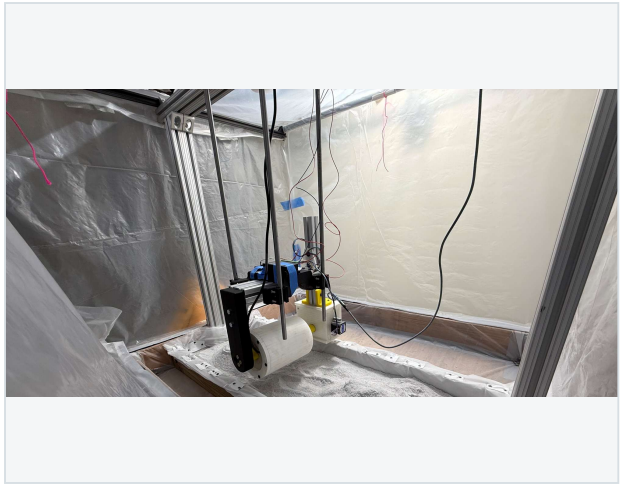
Vibratory Compaction of Lunar Regolith

NASA Nebraska Space Grant Fellowship | Undergraduate Researcher | 2025-2026

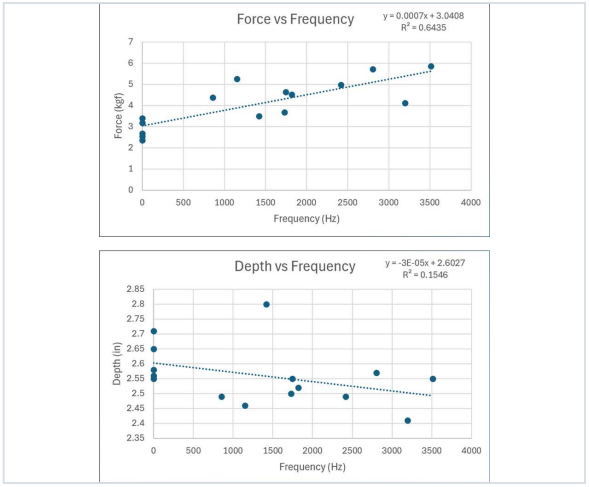
OUTCOME Higher-frequency vibration increased measured penetration resistance by ~65%; ANOVA showed a significant frequency effect on resistance (p = 0.0002) but not compaction depth.



Final roller design - internal vibration motor.



Controlled testing in lunar regolith simulant.



Frequency-response analysis.

OBJECTIVE

Investigate whether wheel-integrated vibration could improve the mechanical response of lunar regolith during rover traversal and support preparation of roads, landing pads and surface infrastructure.

DESIGN & BUILD

Compared three concepts with a weighted trade study, selected an adjustable vibratory roller and completed three design iterations. Integrated an internal vibration motor and adapted the final design to the RIDER Jr. test platform.

EXPERIMENTAL VALIDATION

Compared vibration-off and vibration-on conditions while holding regolith type, density, travel speed, RIDER Jr. and number of passes constant. Measured penetration force and depth across vibration conditions.

7.61 -> 12.58 psi
MEAN STRESS

0.0002
FORCE ANOVA P

0.18
DEPTH ANOVA P

INTERPRETATION
Vibration increased penetration resistance without a statistically significant change in measured depth.

Presented at the Nebraska Academy of Sciences - Aerospace Section.